PRADNYESH DIWAN

8554948282 | pradnyeshdiwan589@gmail.com | linkedin.com/in/pradnyeshddiwan

Education

Vellore Institute of Technology Jan 2023 – Present
B.Tech, Electronics & Communication Engineering; CGPA: 8.94 *Vellore, India*

Anglo Urdu Boys High School Feb 2023
12th Grade (State Board) – 79.2% *Pune, India*

St. Joseph Convent School Aug 2021
10th Grade (CBSE) – 95% *Maharashtra, India*

Experience

Embedded System Design Intern – Maven Silicon May 2025

- Completed hands-on training in embedded systems architecture, hardware selection, firmware coding, and testing across multiple lab-based mini-projects.
- Built a real-time automation system on ESP32 integrating 3 sensors (ultrasonic, LDR, DHT11) for autonomous fan and light control.
- Debugged and validated embedded firmware through iterative test cycles, reinforcing hardware-software integration skills.

Projects

4-Bit Priority Encoder | *Cadence Virtuoso, gpdk90nm, ADE-L* 2025

- Engineered NAND2 and NAND4 logic gates at the transistor level (parallel PMOS pull-up, series NMOS pull-down networks) using the gpdk90nm CMOS technology library in Cadence Virtuoso.
- Integrated individually-verified gate-level cells into a complete priority encoder architecture, applying a modular schematic-design methodology.
- Performed transient simulation and waveform analysis in the ADE-L environment using VPULSE input stimuli, achieving 100% correlation between simulated outputs and the expected truth table across all input conditions.

ESP32-Based Smart Home Automation System | *ESP32, DHT11, Ultrasonic Sensor, LDR, 2-Channel Relay Module, Embedded C/C++* 2025

- Architected a dual-condition decision system on ESP32 that integrates ultrasonic (presence), LDR (ambient light), and DHT11 (temperature) sensor data to independently control fan and light relays.
- Implemented modular Embedded C firmware with dedicated functions for sensor acquisition, threshold-based logic, and relay actuation, enabling real-time processing and easy maintainability.
- Achieved sub-2-second response time from sensor input to relay activation, validated through structured unit, integration, and 6-hour continuous stress testing across all input combinations.

Verilog-Based Adaptive Traffic Light Controller | *Verilog HDL, Multisim* 2025

- Designed a sensor-driven adaptive traffic light controller in Verilog HDL, dynamically granting right-of-way to farm vehicles while preserving highway priority through an always-green default state.
- Modeled multi-state finite state machine (FSM) control logic to arbitrate competing traffic priorities under varying sensor input conditions.
- Simulated and validated circuit-level sensor interfacing and control logic in Multisim to verify correct system behavior before finalizing the design.

Technical Skills

VLSI Design: Cadence Virtuoso (Schematic, ADE-L, Symbol View), CMOS Logic Design, gpdk90nm Library, Transient Analysis
Programming/HDL: Verilog, C , C++ , Embedded C
Tools: MATLAB/Simulink, Multisim, ESP32, Sensor Interfacing (Ultrasonic, LDR, DHT11)